Robustness of AI-Text Detection Under Text Editing and Paraphrasing

1. Real-world problem

Why robustness matters

AI-generated text can be lightly edited or paraphrased before publication.

A detector that performs well on untouched text may rely on surface patterns rather than stable signals.

False positives can wrongly label human writing; false negatives can let AI-generated text pass undetected.

This experiment tests one classical text classifier under controlled text transformations.

3 conditions

Original • Surface edit • Paraphrase

300 pairs

Held-out question topics

600 texts

Per condition: 300 human + 300 AI

2. Research question & hypotheses

Research question

How much does AI-text detection change when AI-generated passages are surface-edited or paraphrased?

H1 — Surface edits

Small wording/formatting changes may have limited effect if learned patterns remain present.

H2 — Semantic paraphrase

Rewriting sentences and structure is expected to reduce recall more than surface editing.

Primary outcome: AI recall. Supporting outcomes: accuracy, precision, F1-score and false positive rate.

3. Dataset & evaluation sample

HC3

Human–ChatGPT Comparison Corpus

5 sources

Finance, Medicine, Open QA, Reddit ELI5, Wiki CSAI

17,528

Training questions

4,383

Held-out questions

Prepared data were deduplicated and filtered to topics containing both human and AI answers.

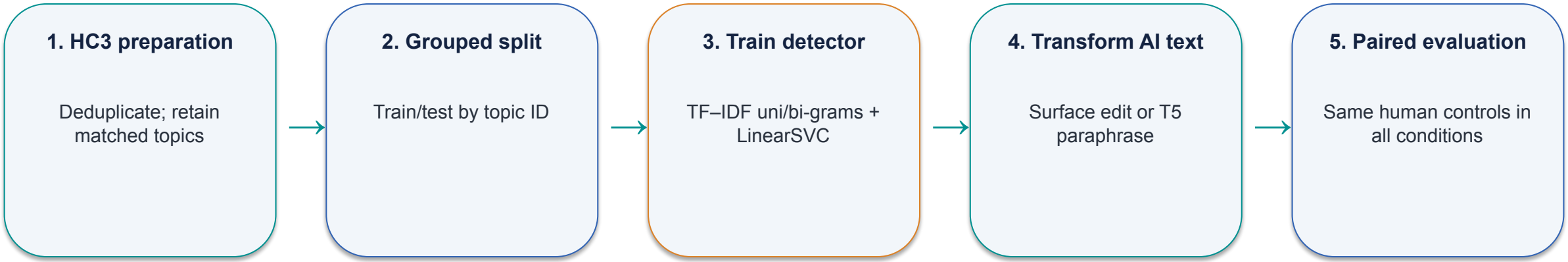
GroupShuffleSplit (test_size=0.20, random_state=42) grouped by topic_id to reduce train/test leakage.

Final robustness sample: 300 held-out topics, each with one human passage and one AI passage.

Each condition contains 600 passages: 300 human controls plus 300 AI texts in that condition.

Source counts are uneven; Reddit ELI5 dominates the paired sample.

4. Methodology & experimental controls



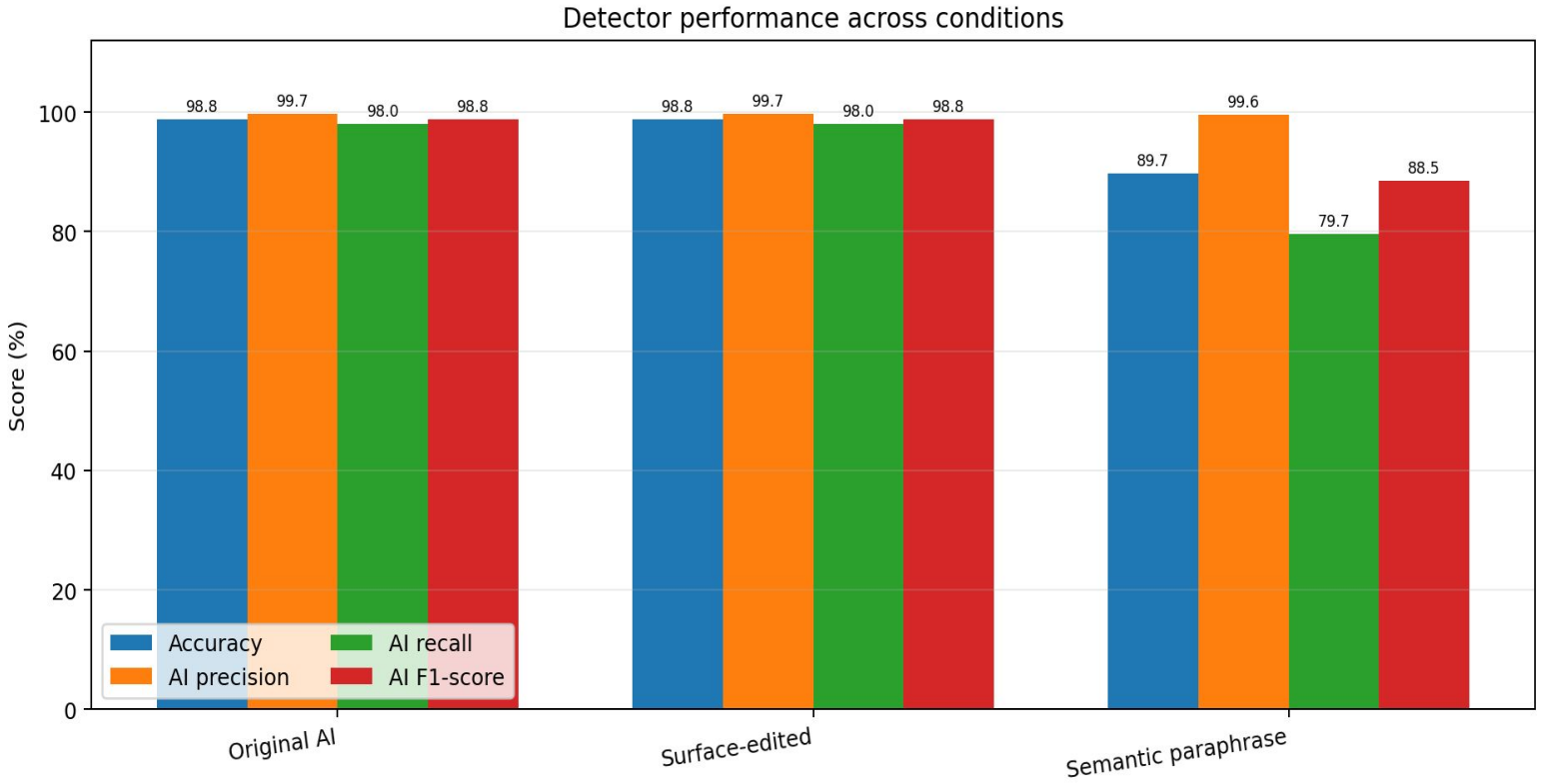
TF–IDF word unigrams/bigrams + LinearSVC, class_weight=balanced.

Train only on original training text; edited/paraphrased text used only for held-out evaluation.

Metrics calculated on combined human + AI passages, not AI-only subsets.

5. Results: overall metrics

Controlled paired comparison — 600 passages per condition



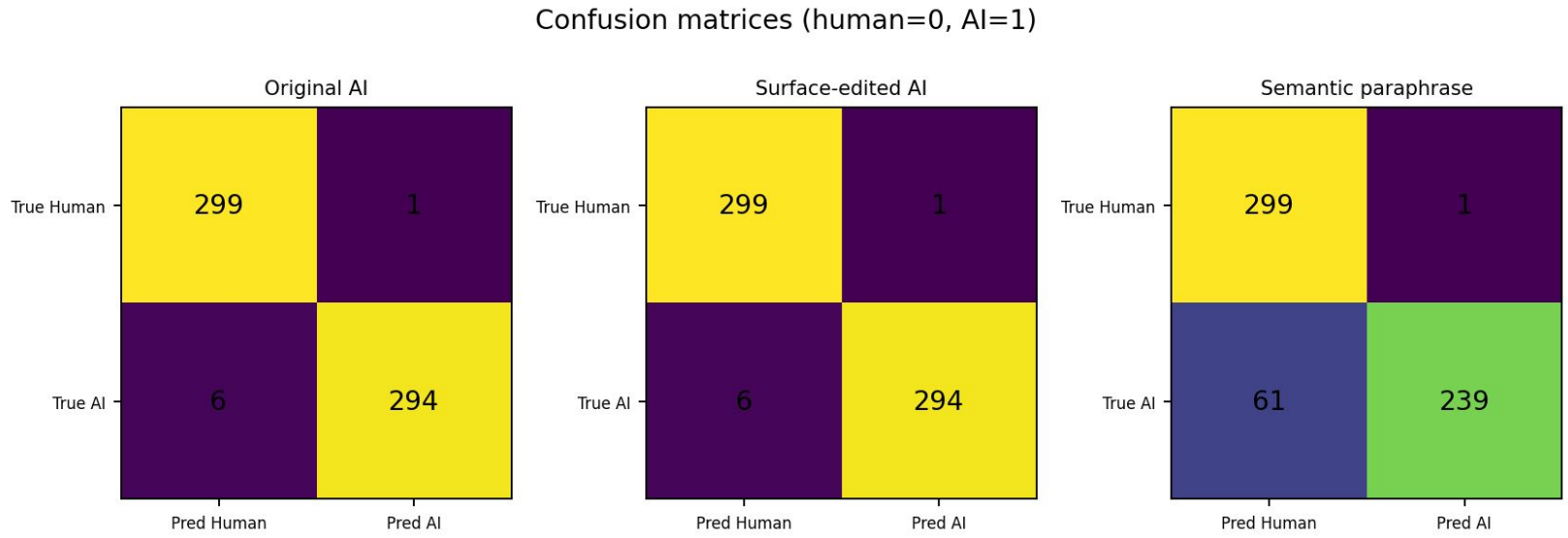
Metric	Original	Surface	Paraphrase
Accuracy	98.83%	98.83%	89.67%
AI recall	98.00%	98.00%	79.67%
AI F1	98.82%	98.82%	88.52%

AI recall change

−18.33 percentage points

FPR: 0.33% in all three conditions. Human false positives: 1/300 per condition.

6. What changed? Confusion matrices



6 → 61

AI passages missed

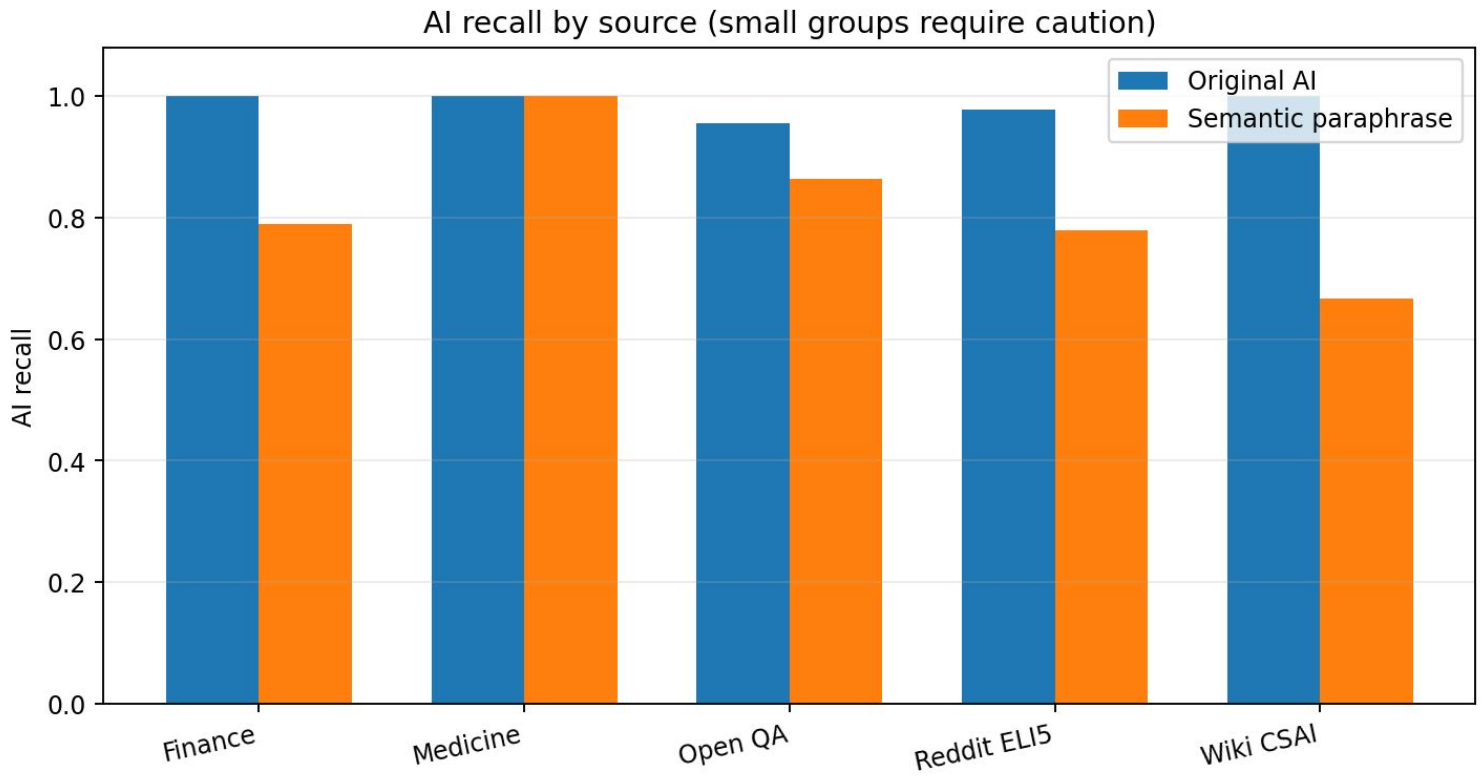
1

Human passages flagged AI

Original and surface-edited counts were identical.

Paraphrasing added 55 false negatives.

7. Source-level results



AI recall (AI passages per source):

Finance (n=38): 100% → 78.95%.

Reddit ELI5 (n=222): 97.75% → 77.93%.

Open QA (n=22): 95.45% → 86.36%.

Medicine (n=15): unchanged in this sample.

Wiki CSAI (n=3): 66.67% after paraphrasing.

Unequal and small groups require caution.

8. Failure cases & limitations

Observed paraphrase quality issues

One output changed the 1 GHz processor example into an incorrect calculation claim.

Another changed an apple example to pencils, creating an inconsistency.

Some outputs removed important qualifications or numerical details.

Study limitations

Only 300 paired held-out questions were evaluated.

Paraphrases were not all meaning-preserving.

Reddit ELI5 dominates; source counts are uneven.

Only one classical detector and one paraphraser were tested.

This measures robustness to generated paraphraser outputs, not exclusively meaning-preserving paraphrases.

9. Conclusion & future work

Conclusion

In this HC3 sample, original AI recall was 98.00%; the tested surface edits did not measurably change performance.

Semantic paraphraser outputs reduced recall to 79.67% (−18.33 pp); false negatives rose from 6 to 61.

This is evidence about this detector, dataset, and paraphrasing process—not all AI detectors or all meaning-preserving rewrites.

Future work

Human-review paraphrases; report raw and quality-filtered results.

Expand and balance the sample; test more detectors and paraphrasers.

Evaluate additional domains, human edits, and model confidence.